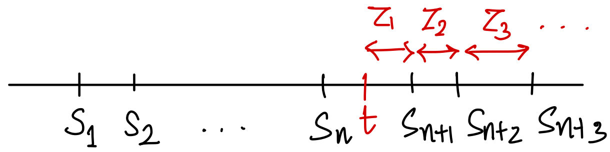
EE2080 : Lecture 7 : 11 Aug 2025

RECAP: * Poisson Processes.

- $Z_1 \sim \exp(\lambda)$
- $Z_1, Z_2, \ldots \overset{iid}{\sim} \exp(\lambda)$
- $N_t \sim \text{Pos}(\lambda t)$.



PLAN: – Discrete Time Markov Chains

- Markov Property, Transition Probability Matrix
- Time Homogeneous DTMC – Chapman Kolmogorov Eqn.

REFERENCE: A. Kumar, "Discrete Event Stochastic Processes"

Conditional Independence.

Given a probability space $(\Omega, \mathcal{F}, \mathbb{P})$, two events A and $B \in \mathcal{F}$ are said to be conditionally independent given another event $C \in \mathcal{F}$ s.t $\mathbb{P}(C) > 0$ $(A \perp\!\!\!\perp B \mid C)$ if $\boxed{\mathbb{P}(A \cap B \mid C) = \mathbb{P}(A \mid C)\, \mathbb{P}(B \mid C)}$ or equivalently $\boxed{\mathbb{P}(A \mid BC) = \mathbb{P}(A \mid C)}$

For $C \in \mathcal{F}$ s.t $\mathbb{P}(C) > 0$, define $\mathbb{P}_C : \mathcal{F} \to [0,1]$ as $\mathbb{P}_C(A) = \mathbb{P}(A \mid C)$ for $A \in \mathcal{F}$

Homework 1) Prove that $\mathbb{P}_C$ is a probability measure on $(\Omega, \mathcal{F})$.

Example: The random variable $X_0 \sim \text{unif}(\{0.1, 0.9\})$. When X_0 takes value $p \in \{0.1, 0.9\}$, X_1 and X_2 are iid $\text{Ber}(p)$ random variables.
Define the events $A = \{X_1 = 1\}$, $B = \{X_2 = 1\}$, $C = \{X_0 = 0.1\}$. Is $A \perp\!\!\!\perp B$? Is $A \perp\!\!\!\perp B \mid C$?

The Markov Property.

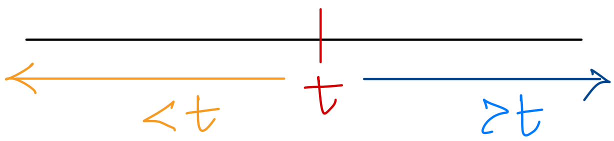
Fix a probability space $(\Omega, \mathcal{F}, \mathbb{P})$

Defn: A process $\{X_t, t \in T\}$ is said to be a Markov process if for any $t \in T$

$$(X_s : s < t) \perp\!\!\!\perp (X_s : s > t) \mid X_t$$

$\forall\, m, n \in \mathbb{N},\ s_1, s_2, \ldots, s_m, t_1, t_2, \ldots, t_n \in T$

s.t. $\underbrace{s_1 < s_2 < \ldots < s_m}_{t-} < t < \underbrace{t_1 < t_2 < \ldots < t_n}_{>t}$



$$\mathbb{P}\Big(\underbrace{X_{s_1} \leq x_1, X_{s_2} \leq x_2, \ldots, X_{s_m} \leq x_m}_{\text{before } t}, \underbrace{X_{t_1} \leq y_1, X_{t_2} \leq y_2, \ldots, X_{t_n} \leq y_n}_{\text{after } t} \Big| X_t \leq x\Big)$$

$$= \mathbb{P}\left(X_{s_1} \leq x_1, X_{s_2} \leq x_2, \ldots, X_{s_m} \leq x_m \,\middle|\, X_t \leq x\right) \mathbb{P}\left(X_{t_1} \leq y_1, X_{t_2} \leq y_2, \ldots, X_{t_n} \leq y_n \,\middle|\, X_t \leq x\right)$$

> Given the present, the future is independent of the past.

Discrete time Markov Chain : Fix a probability space $(\Omega, \mathcal{F}, \mathbb{P})$

Defn: A stochastic process $\{X_n, n \in \mathbb{N}\}$ taking values in a countable set S is called a discrete time Markov chain (DTMC) if it has the Markov property, i.e, for all $i_1, i_2, \ldots, i_{n-1}, i_n = i, j \in S$ and $n \in \mathbb{N}$,

$$\mathbb{P}\{X_{n+1} = j \mid X_1 = i_1, \ldots, X_{n-1} = i_{n-1}, X_n = i\} = \mathbb{P}\{X_{n+1} = j \mid X_n = i\}$$

At time n, X_n is the present, $X_1, \ldots, X_{n-1}$ form the past and X_{n+1} is the future.

The term "chain" comes from the fact that $\{X_n, n \in \mathbb{N}\}$ takes values in a countable set S. The values taken by the process are called states of the process. The set S is called the state space.

Example: Consider repeated tosses of a coin with prob. of heads being p.

$$X_n = \begin{cases} 1 & \text{if } n^{th} \text{ toss is } H \\ 0 & \text{if } n^{th} \text{ toss is } T \end{cases}, \quad n \geq 1.$$

$X_1, X_2, \ldots$ iid $\therefore$ Markov Property holds trivially.

Define $Y_n = \sum_{i=1}^{n} X_i$, $n \geq 1$ $\quad$ Y_i – # heads till n^{th} toss., $Y_n \in \mathbb{Z}^+$

$$\mathbb{P}\{Y_{n+1} = j \mid Y_1 = i_1, Y_2 = i_2, \ldots, Y_n = i\} = \begin{cases} 0 & \text{if } j \notin \{i, i+1\} \\ p & \text{if } j = i+1 \\ 1-p & \text{if } j = i \end{cases}$$

$$= \mathbb{P}(Y_{n+1} = j \mid Y_n = i)$$

Thus $\{Y_n, n \in \mathbb{N}\}$ is a DTMC on $\mathbb{Z}^+$.

Note: $\{Y_n, n \in \mathbb{N}\}$ is a DTMC $\Rightarrow (Y_{n+1} \perp\!\!\!\perp (Y_1, \ldots Y_{n-1}) \mid Y_n)$

$$\not\Rightarrow Y_{n+1} \perp\!\!\!\perp (Y_1, \ldots Y_{n-1})$$

Unconditional independence is not implied !!

? Are Y_{n+1} and Y_{n-1} independent unconditionally in the above example ?

Homework: 1) If $\{X_n, n \in \mathbb{N}\}$ is a DTMC on S, then for any $A_1, A_2, \ldots A_{n-1} \subset S$ and $i, j \in S$, prove that

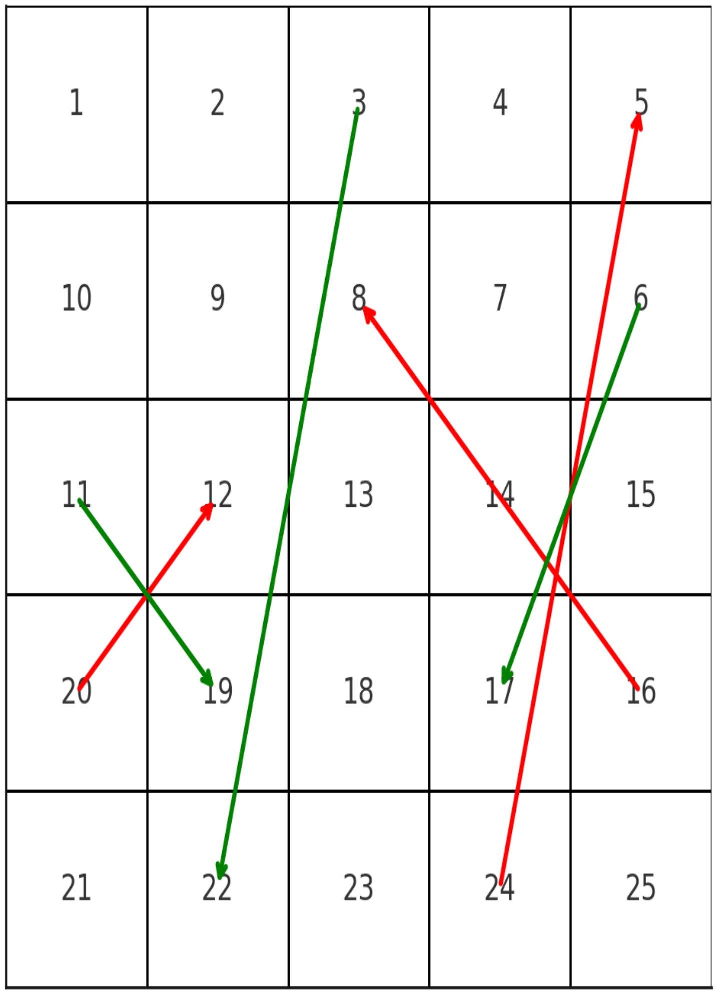
$$\mathbb{P}(X_{n+1} = j \mid X_1 \in A_1, X_2 \in A_2, \ldots, X_{n-1} \in A_{n-1}, X_n = i) = \mathbb{P}(X_{n+1} = j \mid X_n = i)$$

2) If $\{X_n, n \in \mathbb{N}\}$ is a DTMC on S, then for $n < n_1 < n_2 < \cdots < n_m$, and $i_1, i_2, \ldots, i_{n-1}, i, j_1, \ldots, j_m \in S$, Show that.

$$\mathbb{P}\{X_{n_1} = j_1, X_{n_2} = j_2, \ldots, X_{n_m} = j_m \mid X_1 = i_1, X_2 = i_2, \ldots, X_n = i\} =$$
$$\mathbb{P}\{X_{n_1} = j_1, X_{n_2} = j_2, \ldots, X_{n_m} = j_m \mid X_n = i\}.$$

Remark: In the defn of DTMC, the conditional independence of the immediate next step after n is only given. The above result $\Rightarrow$ the joint conditional independence of any finite subset of random variables in the future of the past given the present !!

5x5 Snakes and Ladders



The state space $S = \{1, 2, 3, \ldots, 25\}$

Assume $X_n = 4$ at some time n.

What are the values that X_{n+1} can take?

$X_{n+1} \in \{5, 17, 7, 8, 9, 10\}$

For all $j, i_1, i_2, \ldots, i_{n-1} \in S$, we have

$$P(X_{n+1} = j \mid X_n = 4, X_{n-1} = i_{n-1}, \ldots, X_1 = i_1)$$
$$= P(X_{n+1} = j \mid X_n = 4)$$

Suffices to retain the most recent information and discard the history!!

Transition Probability Matrix

Let $\{X_n, n \geq 0\}$ be a DTMC with state space S.

The first order pmf of X_n is denoted as $\pi_n = (\pi_i(n), i \in S)$.

i.e $\pi_i(n) = \mathbb{P}\{X_n = i\}$.

The conditional probabilities of the form $\mathbb{P}\{X_{n+1} = j \mid X_n = i\}$ are denoted as $P_{i,j}(n)$ and are called transition probabilities at time step n.

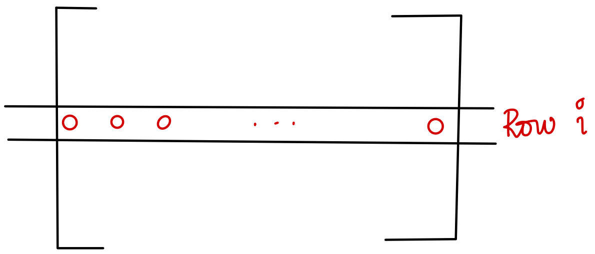
time n time $n+1$

$$i \xrightarrow{P_{i,j}(n)} j$$

Defn: Let $\{X_n, n \in \mathbb{N}\}$ be a DTMC with state space S. The transition probability matrix (TPM) of the Markov chain at any time $n \in \mathbb{N}$ is a $|S| \times |S|$ matrix

$P(n) = [P_{i,j}(n)]_{i,j \in S}$ defined as $P_{i,j}(n) = \mathbb{P}\{X_{n+1} = j \mid X_n = i\}$, $i, j \in S$

Transition Probability Matrix



For each $i \in S$, $\sum_{j \in S} P_{i,j}(n) = 1$

A matrix with non-negative entries and row sums equal to 1 is called a row-stochastic matrix

$P(n)$ is a row stochastic matrix for every n.

Row i of $P(n)$ = Conditional PMF of X_{n+1} given $X_n = i$. ($\because$ each row is a PMF on S).

$$\sum_{j \in S} j\, P_{i,j}(n) = \mathbb{E}\left[X_{n+1} \,\middle|\, X_n = i\right]$$

A TPM $P(n)$.

$$\underbrace{\begin{bmatrix} & & \\ & & \\ & & \end{bmatrix}}_{P(n)} \begin{bmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix} = ?$$

$|S| \times |S|$

Time-Homogeneous DTMC

Defn: A DTMC $\{X_n, n \in \mathbb{N}\}$ with state space S and TPMs $\{P(n)\}_{n=1}^{\infty}$ is said to be time homogeneous if $P(n) = P(n+1)$, $\forall n \in \mathbb{N}$

In this case, we denote the common TPM as P.

$$\text{i.e. } \mathbb{P}\left(X_{n+1} = j \,\middle|\, X_n = i\right) = \mathbb{P}\left(X_2 = j \,\middle|\, X_1 = i\right).$$

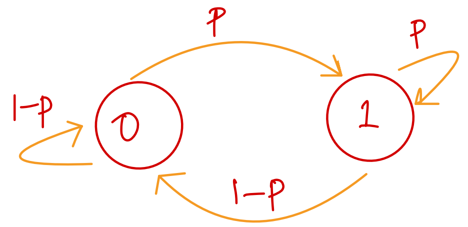
Example: $\{X_n\}_{n=1}^{\infty}$ iid $Ber(p)$ $\qquad X_i = \begin{cases} 1 & \text{w.p } p \\ 0 & \text{w.p } 1-p. \end{cases}$ $\qquad S = \{0, 1\}$

$$\text{TPM, } P = \begin{array}{c} \\ 0 \\ 1 \end{array} \begin{array}{c} \begin{array}{cc} 0 & 1 \end{array} \\ \begin{bmatrix} 1-p & p \\ 1-p & p \end{bmatrix} \end{array}$$

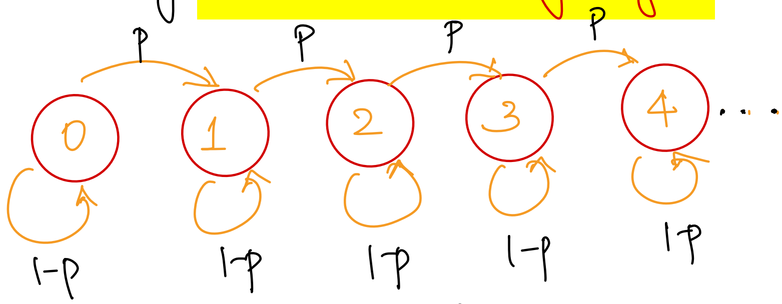
$$Y_n = \sum_{i=1}^{n} X_i \qquad S = \mathbb{N} \cup \{0\}$$

$$\begin{array}{c|cccccc} & 0 & 1 & 2 & 3 & \cdots \\ \hline 0 & 1-p & p & 0 & 0 & \cdots \\ 1 & 0 & 1-p & p & 0 & \cdots \\ 2 & 0 & 0 & 1-p & p & 0 \cdots \\ \vdots & & & & & \end{array}$$

Sometimes, transition probabilities are described using transition probability diagram



$\{X_n, n \in \mathbb{N}\}$



$\{Y_n, n \in \mathbb{N}\}$

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n-step transition probabilities $\mathbb{P}\{X_{n+1} = j \mid X_1 = i\} = p_{i,j}^{(n)}$

i.e given that you are in state i, what is the prob. that you will be in state j after n time steps?

This is talked about only for time-homogeneous DTMCs.

Consider $\{X_n, n \in \mathbb{N}\}$ DTMC with state space S.

$\{X_n, n \in \mathbb{N}\}$ is Time Homogeneous with TPM P.

$$\mathbb{P}\{X_{n+1} = j \mid X_n = i\} = P_{i,j} \quad \forall n \in \mathbb{N}$$

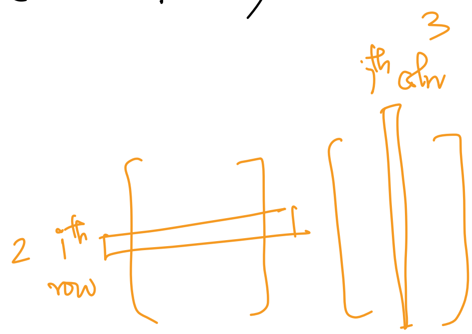
$$\mathbb{P}\{X_2 = j \mid X_1 = i\} = P_{i,j}$$

$$\mathbb{P}\{X_3 = j \mid X_1 = i\} = \sum_{k \in S} \mathbb{P}(X_3 = j \mid X_2 = k, X_1 = i)\, \mathbb{P}(X_2 = k \mid X_1 = i)$$

$$= \sum_{k \in S} \mathbb{P}(X_3 = j \mid X_2 = k)\, \mathbb{P}(X_2 = k \mid X_1 = i)$$

$$= \sum_{k \in S} P_{k,j} \cdot P_{i,k}$$

$$= (P^2)_{i,j}$$



We can get two-step transition probabilities from the one-step TPM!!